Experimental Proposal -- Confirmation of a Dielectric Longitudinal Delay of a Bright Interference Fringe

A dielectric-first derivation and time-of-flight test in a Mach-Zehnder interferometer

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One-Sentence Summary. A bright interference fringe produced at Mach-Zehnder recombination carries increased constructive-interference energy density and therefore exhibits a dielectric longitudinal delay relative to an ordinary reference beam.

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1. Experimental Proposal – Confirmation of a Dielectric Longitudinal Delay of a Bright Interference Fringe

1.1. Abstract

Light propagates more slowly in a transparent dielectric because the electric and magnetic response of the medium increases the effective permittivity and permeability. In the usual electromagnetic description, the propagation speed therefore becomes

$$c_{\text{eff}} = \frac{1}{\sqrt{\varepsilon_{\text{eff}}\mu_{\text{eff}}}} = \frac{c}{n}.$$

The same kind of slowdown follows when two coherent beams are brought into coherent superposition in the recombination phase of a Mach-Zehnder interferometer. The first beam splitter only prepares two bright arm beams. The relevant step appears later, when those beams recombine. These are not the same spatial operation run backward (see Section 4).

If each beam is treated as the coherent electromagnetic response seen by the other, then equal-beam constructive interference doubles the fields, quadruples the instantaneous energy density, and gives the same reduced-speed result

$$c_{\text{eff}} = \frac{c}{2}.$$

The standard output reading instead assigns the bright output density $2u_0$ and therefore gives no anomalous delay. The experiment is a direct time-of-flight test between these two readings.

1.2. 1. Rationale

In a linear dielectric, an incident electromagnetic wave induces a response electromagnetic wave in the medium. Maxwell electrodynamics is linear, so the physical field is the sum of the incident and response sectors.

When that response is in phase with the incident wave, the fields add, the local electromagnetic energy density increases, and the propagation speed is reduced through the larger effective permittivity and permeability.

Constructive interference has the same local structure. Two coherent beams add linearly in phase to form a larger total field. At the level of the electromagnetic field equations, that is the same loading pattern: one sector is carried together with a second in-phase response sector.

Dielectric slowing follows from this linear in-phase field addition in electromagnetism. The same dielectric-like longitudinal delay therefore follows at constructive interference.

This logic is one-sided. Constructive interference yields a denser surviving bright fringe, while destructive interference depletes the field and in the dark-limit cancels the fringe rather than producing a faster propagated one.

1.3. 2. Classic Dielectric Result

In a transparent linear dielectric,

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P},$$

and

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M}).$$

For a linear response,

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}.$$

and

$$\mathbf{M} = \chi_m \mathbf{H}.$$

Therefore

$$\mathbf{D} = \varepsilon_0 (1 + \chi_e) \mathbf{E} = \varepsilon_{\text{eff}} \mathbf{E},$$

so

$$\varepsilon_{\text{eff}} = \varepsilon_0 (1 + \chi_e).$$

If the magnetic sector is likewise loaded,

$$\mu_{\text{eff}} = \mu_0 (1 + \chi_m).$$

In a source-free region,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}.$$

Substituting the effective constitutive relations gives

$$\nabla \times \mathbf{E} = -\mu_{\text{eff}} \frac{\partial \mathbf{H}}{\partial t}, \qquad \nabla \times \mathbf{H} = \varepsilon_{\text{eff}} \frac{\partial \mathbf{E}}{\partial t}.$$

Taking the curl of the first equation, and using the second together with $\nabla \cdot \mathbf{E} = 0$, gives the wave equation

$$\nabla^2 \mathbf{E} - \varepsilon_{\text{eff}} \mu_{\text{eff}} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0.$$

Therefore the propagation speed is

$$c_{\text{eff}} = \frac{1}{\sqrt{\varepsilon_{\text{eff}} \mu_{\text{eff}}}}.$$

For symmetric electric and magnetic loading,

$$\chi_e = \chi_m = k,$$

so

$$\varepsilon_{\text{eff}} = \varepsilon_0 (1 + k), \qquad \mu_{\text{eff}} = \mu_0 (1 + k),$$

and therefore

$$c_{\text{eff}} = \frac{1}{\sqrt{\varepsilon_0 \mu_0 (1 + k)^2}} = \frac{c}{1 + k}.$$

This is the standard reduced-speed result.

1.4. 3. Same Structure at Coherent Superposition

The same electromagnetic loading structure appears when two coherent beams are coherently superposed in vacuum during recombination.

Let one beam have fields

$$\mathbf{E}_1, \qquad \mathbf{H}_1,$$

and let the second beam act as a coherent response field proportional to the first:

$$\mathbf{E}_2 = k \mathbf{E}_1, \qquad \mathbf{H}_2 = k \mathbf{H}_1.$$

Here the coherent superposition is described in the same form as the dielectric case, but with the second coherent beam supplying the response sector. Define, abbreviating *coherent* by the subscript coh,

$$\mathbf{D}_{\text{coh}} = \varepsilon_0(\mathbf{E}_1 + \mathbf{E}_2), \qquad \mathbf{B}_{\text{coh}} = \mu_0(\mathbf{H}_1 + \mathbf{H}_2).$$

Then

$$\mathbf{D}_{\text{coh}} = \varepsilon_0(1+k)\mathbf{E}_1 = \varepsilon_{\text{coh}}\mathbf{E}_1,$$

with

$$\varepsilon_{\text{coh}} = \varepsilon_0(1+k),$$

and likewise

$$\mathbf{B}_{\text{coh}} = \mu_0(1+k)\mathbf{H}_1 = \mu_{\text{coh}}\mathbf{H}_1,$$

with

$$\mu_{\text{coh}} = \mu_0(1+k).$$

The same source-free Maxwell form then becomes

$$\nabla \times \mathbf{E}_1 = -\frac{\partial \mathbf{B}_{\text{coh}}}{\partial t}, \qquad \nabla \times \mathbf{H}_1 = \frac{\partial \mathbf{D}_{\text{coh}}}{\partial t}.$$

So

$$\nabla \times \mathbf{E}_1 = -\mu_{\text{coh}}\frac{\partial \mathbf{H}_1}{\partial t}, \qquad \nabla \times \mathbf{H}_1 = \varepsilon_{\text{coh}}\frac{\partial \mathbf{E}_1}{\partial t}.$$

Taking the curl again gives

$$\nabla^2\mathbf{E}_1 - \varepsilon_{\text{coh}}\mu_{\text{coh}}\frac{\partial^2 \mathbf{E}_1}{\partial t^2} = 0,$$

therefore

$$c_{\text{eff}} = \frac{1}{\sqrt{\varepsilon_{\text{coh}}\mu_{\text{coh}}}} = \frac{1}{\sqrt{\varepsilon_0\mu_0(1+k)^2}} = \frac{c}{1+k}.$$

For equal beams at constructive interference,

$$k = 1,$$

so

$$c_{\text{eff}} = \frac{c}{2}.$$

At the bright center of the fringe, where the interference is constructive, the total fields are

$$\mathbf{E}_{\text{tot}} = 2\mathbf{E}_1, \qquad \mathbf{H}_{\text{tot}} = 2\mathbf{H}_1,$$

and the instantaneous energy density there is

$$u_{\text{tot}} = 4u_1.$$

More generally, across the fringe profile one may write

$$u(x) = 4u_1 \cos^2\left(\frac{\Delta\phi(x)}{2}\right),$$

so the factor 4 is the bright-center value of the full spatial dependence.

That is the tested coherent-overlap result.

1.5. 4. Why the Split Phase Is Different

A Mach-Zehnder has two distinct phases.

At the first beam splitter:

- one bright incident beam becomes two bright arm beams
- the purpose is to prepare coherent arm beams
- no loaded constructive-interference fringe is being claimed

At recombination:

- two arm beams meet coherently
- the fields add or subtract before the ordinary output decomposition is written
- the loaded-fringe question is whether the bright raw overlap itself propagates with the reduced speed above

Mathematically, the same 50/50 beam-splitter matrix can describe both interfaces, but the occupied input states are different. If

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix},$$

then the split phase starts from one populated input port,

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} A \\ 0 \end{pmatrix},$$

so

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} A/\sqrt{2} \\ A/\sqrt{2} \end{pmatrix},$$

which gives two bright arm beams. Recombination starts instead from two populated arm inputs,

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} A \\ A \end{pmatrix},$$

so

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} \sqrt{2}\,A \\ 0 \end{pmatrix},$$

which gives one bright output and one dark output. The optical element is the same, but the populated states are different, so the split and recombination phases are not the same spatial decomposition.

This also explains why recombination still produces two spatially distributed output arms rather than a single merged beam. The recombiner is a two-port linear map: it sends the two incoming arm fields into the sum and difference output channels. Those channels emerge in different spatial directions, so the recovered field is written across two output arms even when one of them is dark at a bright point.

So the slowdown question belongs to recombination, not to the initial split.

1.6. 5. Experimental Fork

At a "lossless" 50/50 recombiner, the ordinary output fields are

$$\mathbf{E}_+ = \frac{\mathbf{E}_1 + \mathbf{E}_2}{\sqrt{2}}, \qquad \mathbf{E}_- = \frac{\mathbf{E}_1 - \mathbf{E}_2}{\sqrt{2}},$$

and likewise for the magnetic fields.

Here "lossless" means the splitter is modeled as conserving total optical energy, so all energy is accounted for across the two output arms. Small real losses do not change the core result being tested, and in a well-aligned setup they are negligible compared with the claimed factor-of-two speed difference.

The factor $1/\sqrt{2}$ appears because the splitter divides the incoming energy into two equal beams. Energy density is a positive quadratic quantity in the field *amplitudes*, so halving the energy means scaling the field amplitudes by $1/\sqrt{2}$, not by $1/2$. The same factor is written on both the electric and magnetic fields because they are modeled as vector fields and enter the electromagnetic energy density quadratically.

For equal beams at a bright output point, the standard output reading gives

$$u_+ = 2u_0, \qquad u_- = 0.$$

So the experimental fork is:

- ordinary output reading:
$$v_{\text{bright}} = v_{\text{ref}}$$
- loaded raw-overlap reading:
$$v_{\text{bright}} = \frac{v_{\text{ref}}}{2}$$
in the equal-beam limit

1.7. 6. Experimental Idea

Use a coherent source, modulate it, split it into two arms, and recombine the arms so they form stable fringes. Then:

1. isolate one bright interference fringe with an aperture
2. propagate that selected fringe over distance L
3. propagate a matched reference beam over the same distance
4. compare delay slopes $d\tau/dL$

The standard output reading predicts equal slopes. The loaded-fringe reading predicts a larger slope for the bright fringe.

For the equal-beam limit,

$$\tau_{\text{bright}} - \tau_{\text{ref}} \approx \frac{L}{c}.$$

So at 1 m the extra delay is about 3.34 ns, and at 10 m it is about 33.4 ns.

1.8. 7. Detailed Derivation

1.8.1. 7.1 Dielectric Derivation

Start from

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}, \qquad \mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}.$$

Then

$$\varepsilon_{\text{eff}} = \varepsilon_0 (1 + \chi_e).$$

If the magnetic sector is likewise loaded,

$$\mu_{\text{eff}} = \mu_0 (1 + \chi_m).$$

So

$$c_{\text{eff}} = \frac{1}{\sqrt{\varepsilon_{\text{eff}} \mu_{\text{eff}}}}.$$

For symmetric loading $\chi_e = \chi_m = k$,

$$c_{\text{eff}} = \frac{c}{1 + k}.$$

1.8.2. 7.2 Instantaneous Constructive-Interference Derivation

No time averaging is needed for the basic bright-point argument.

At a full constructive-interference point for equal beams,

$$\mathbf{E}_1(t) = \mathbf{E}_0(t), \qquad \mathbf{E}_2(t) = \mathbf{E}_0(t),$$

and

$$\mathbf{B}_1(t) = \mathbf{B}_0(t), \qquad \mathbf{B}_2(t) = \mathbf{B}_0(t).$$

Then

$$\mathbf{E}_{\text{tot}}(t) = 2\mathbf{E}_0(t), \qquad \mathbf{B}_{\text{tot}}(t) = 2\mathbf{B}_0(t).$$

The instantaneous electromagnetic energy density is

$$u_{\text{tot}}(t) = \frac{\varepsilon_0}{2}|\mathbf{E}_{\text{tot}}(t)|^2 + \frac{1}{2\mu_0}|\mathbf{B}_{\text{tot}}(t)|^2.$$

Substituting the doubled fields gives

$$u_{\text{tot}}(t) = 4u_0(t).$$

So the bright-point factor of four is already an instantaneous result.

1.8.3. 7.3 Coherent-Response Form

Write the second beam as a coherent response field proportional to the first:

$$\mathbf{E}_2 = k\mathbf{E}_1, \qquad \mathbf{B}_2 = k\mathbf{B}_1.$$

Then

$$\mathbf{E}_{\text{tot}} = (1+k)\mathbf{E}_1, \qquad \mathbf{B}_{\text{tot}} = (1+k)\mathbf{B}_1,$$

so

$$u_{\text{tot}} = (1+k)^2 u_1.$$

Applying the same reduced-speed structure as in the dielectric case gives

$$c_{\text{eff}} = \frac{c}{1+k}.$$

For equal beams at constructive interference, $k = 1$, hence

$$c_{\text{eff}} = \frac{c}{2}.$$

1.9. 8. What the Experiment Decides

This experiment tests which object should be treated as the propagating fringe after recombination:

- the ordinary normalized output mode
- or the isolated raw constructive-interference fringe

If the measured delay matches the reference, the ordinary reading wins. If the measured delay approaches the $c/2$ prediction in the equal-beam limit, the loaded-fringe reading is supported.